The effect of plough agriculture on gender roles: A machine learning approach by Baiardi & Naghi (2024) revisiting Alesina et al. (2013)

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Outline

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Why do gender norms persist?

- **Global Disparities:** Beliefs about the appropriate role of women vary significantly across regions, creating visible differences in female labor force participation (FLFP).
- **Role of Culture:**
  - Economic choices (e.g., labor supply) are not just driven by current incentives but are partially explained by **culture and norms** (Fernandez, 2007; Fernández & Fogli, 2009).
- **Why it Matters (Policy):**
  - Investigating the *origins* of these norms is crucial to understanding these differences.
  - If norms are historically rooted, standard economic policies may be insufficient to reduce inequality (Hansen et al., 2015).

Boserup's Hypothesis and the Research Question

- **Boserup's Hypothesis (1970):** Gender roles originate in pre-industrial agricultural practices.
  - **Shifting Cultivation:**
    - Uses hoe/digging stick; high need for weeding (women's traditional role), facilitating female participation.
  - **Plough Cultivation:**
    - Requires considerable physical strength, giving men an advantage.
- **The Legacy:**
  - Where the plough was used, a gendered division of labor emerged.
  - This persisted as a cultural norm regarding women's roles, continuing to the present day.

Research Question

Does the historical adoption of the plough have a persistent, causal negative effect on female labor force participation today?

Original Econometric Model (Alesina et al., 2013)

- Baiardi & Naghi (2024) replicate and extend the country-level analysis of Alesina et al. (2013).
- **Baseline OLS Regression:**

$$y_c = \alpha + \beta \cdot \text{plough use}_c + X_c^H \Gamma + X_c^C \Pi + \epsilon_c$$

- **Variables:**
  - y_c: Gender role outcomes (e.g. FLFP in 2000).
  - plough use$_c$: Estimated proportion of citizens with ancestors who used the plough in pre-industrial agriculture.
  - X_c^H (**Historical Controls**): Ethnographic variables including ancestral domestication of large animals and ancestral settlement patterns.
  - X_c^C (**Contemporary Controls**): Natural log of real per capita GDP and its square (to capture the U-shaped relationship with economic development).

The Problem: Endogeneity in OLS

1. Reverse Causality

- Plough agriculture may have been more common in countries that already had **less equal gender-role attitudes**.
  $\Rightarrow$ bias OLS away from zero.

2. Omitted Variable Bias

- Plough adoption correlates with **higher historical development**; since development generally increases female participation, omitting this factor masks the plough's negative impact.
  $\Rightarrow$ bias OLS towards zero.

The Solutions (Alesina et al., 2013)

Solution 1: Adding Observables

- Adds two groups of potential confounders: **historical characteristics** of the ancestors of the current population and **current economic, social, and cultural characteristics** of the country.

Solution 2: Instrumental Variables

- Exploits the fact that plough adoption is correlated with land suitability for specific crops.
- **Instruments:** Suitability for **plough-positive** crops (e.g., wheat, barley) and **plough-negative** crops (e.g., maize, sorghum).
- *Exclusion Restriction:* The assumption is that crop suitability affects gender norms *only* through the historical adoption of the plough.

Methodology

Why Use Causal Machine Learning (CML)?

Unknown Functional Forms:

- Theory offers limited guidance on how ancestral variables (e.g., political complexity) interact with current norms.
- Unlike OLS, CML flexibly captures these non-linear nuisance functions.

The "Curse of Dimensionality":

- Alesina et al. (2013) could not include all potential interactions (p) given the small sample size ($N \approx 160$).
- CML handles high-dimensional controls ($p > n$) via regularization, allowing the inclusion of all raw covariates at once.

Systematic Robustness:

- Replaces "ad-hoc" variable selection with systematic, data-driven algorithms to identify the most relevant confounders.

Double Machine Learning (DML)

- Baiardi & Naghi (2024) employ the **Partially Linear Regression Model**:

$$Y = D\theta_0 + g_0(X) + U, \quad \mathbb{E}[U|X, D] = 0$$
$$D = m_0(X) + V, \quad \mathbb{E}[V|X] = 0$$

- **Variables:** Y is the outcome (FLFP), D is the treatment (Plough Use), and X is the high-dimensional set of controls.

- **Estimation Strategy ("Partialling Out"):**
  1. Predict Outcome Y from X using ML $\rightarrow$ Get Residuals ($Y - \mathbb{E}[Y|X]$).
  2. Predict Treatment D from X using ML $\rightarrow$ Get Residuals ($D - \mathbb{E}[D|X]$).
  3. Regress residuals on residuals to obtain θ_0 (the Average Treatment Effect).

- **Intuition:** This removes the confounding effect of X from both the outcome and the treatment, isolating the "clean" variation in plough use.

DML with IV (DML-IV)

- The standard DML model assumes all confounders are observed in X.
- To account for *unobserved* confounders, they extend the framework to include two Instrumental Variables $Z_k = (Z_1, Z_2)$.
- **The Partially Linear IV Model (PLIV):**
  - We maintain the outcome equation but add an instrument Z (Plough Suitability):

$$
\begin{aligned}
Y &= D\theta_0 + g_0(X) + \zeta, \quad \mathbb{E}[\zeta|Z, X] = 0 \\
D &= r_0(X) + T, \quad \mathbb{E}[T|X] = 0 \\
Z_k &= m_k(X) + V_k, \quad \mathbb{E}[V_k|X] = 0 \quad \text{for } k = 1, 2
\end{aligned}
$$

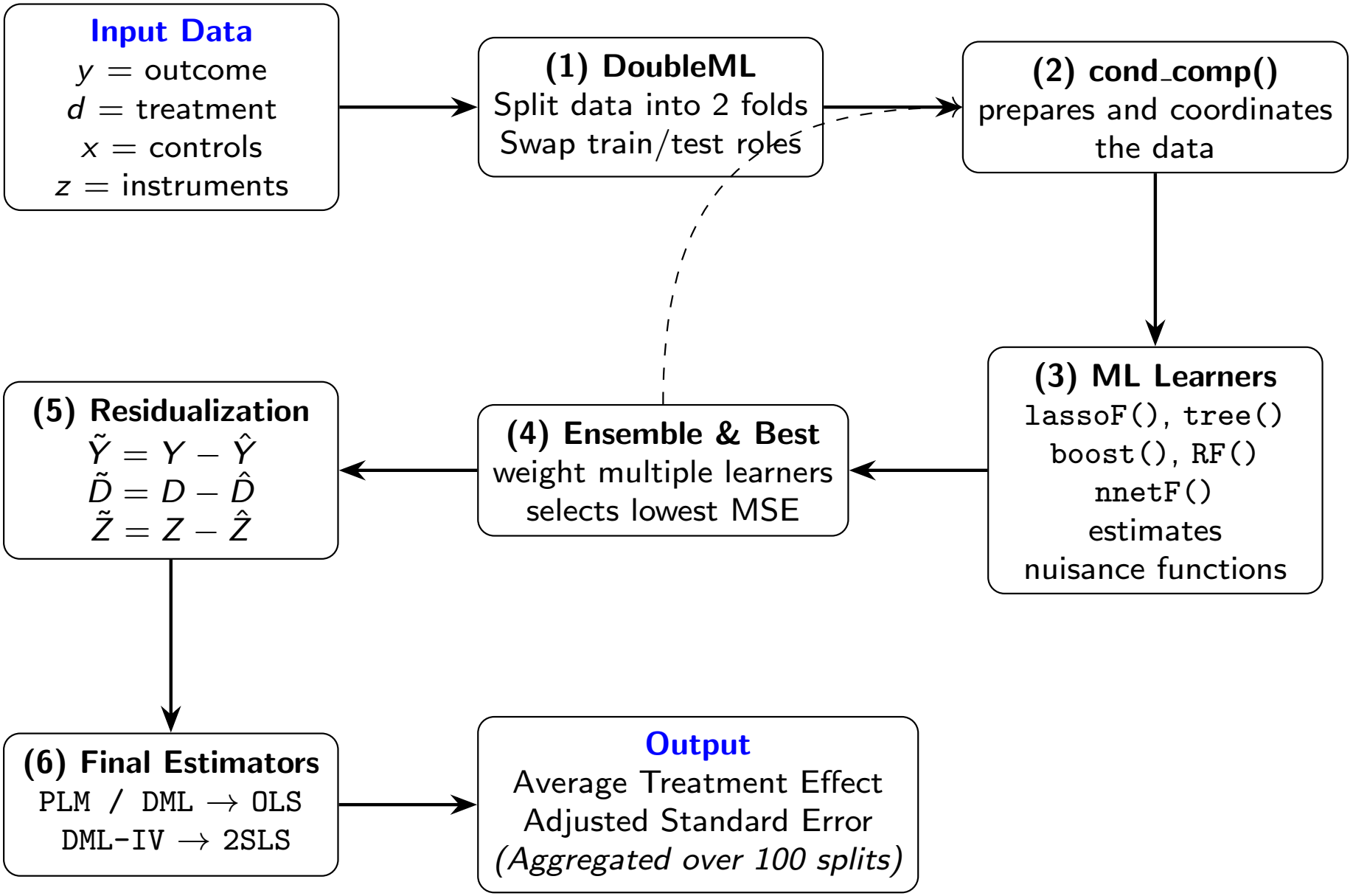
- **Estimation Strategy (3-Step Partialling Out):**
  1. **Prediction:** Predict Y, D, **and** Z using X (via ML algorithms).
  2. **Residualization:** Compute residuals for all three:

$$\tilde{Y} = Y - \hat{Y}, \quad \tilde{D} = D - \hat{D}, \quad \tilde{Z} = Z - \hat{Z}$$

  3. **Estimation:** Estimate θ_0 via Two-Stage Least Squares (TSLS) on the residuals (regressing $\tilde{Y}$ on $\tilde{D}$ using the matrix $\tilde{Z}$ as instruments).

Implementation

Implementation: Pipeline



Implementation Pipeline: DoubleML Functions (1)

Two potential ways of implementing DML:

1. `DoubleML` **R package:** The standard package setup lacks native support for *multiple* instrumental variables simultaneously.
2. **Customized Functions:** These manually program the Two-Stage Least Squares (2SLS) matrix operations to blend the instruments and extract the causal effect.

Custom DML-IV Matrix Algebra

Let $R = \tilde{Y}$, $T = \tilde{D}$, and $V = \tilde{Z} = [\tilde{Z}_1, \tilde{Z}_2]$, an $n \times 2$ matrix.

First Stage (Optimal Instrument Weighting): Regress treatment on instruments to get the predicted treatment:

$$\hat{T} = V(V'V)^{-1}V'T$$

Second Stage (Causal Estimation): Project the outcome residuals onto the predicted treatment:

$$\hat{\theta}_0 = (\hat{T}'\hat{T})^{-1}\hat{T}'R$$

Implementation Pipeline: The ML Engine (2)(3)(4)

Objective: Generate out-of-sample predictions $(\hat{Y}, \hat{D}, \hat{Z})$ for the held-out fold to strip away the confounding effects of the control variables.

(2) The Wrapper `cond_comp()`**:** A custom R function that preps the data and simultaneously feeds it into a battery of different algorithms.

(3) The ML Learners: The code trains five separate models.

- **Lasso & Trees:** Tuned dynamically using 10-fold Cross-Validation.
- **Random Forest & Boosting:** Hard-coded to grow 1,000 trees.
- **Neural Networks:** Configured with 1 hidden layer and 2 neurons.

(4) Hybrid Methods- Ensemble & Best: Evaluates the out-of-sample MSE of all the learners.

- **Best:** selects the learner that best estimates the nuisance functions.
- **Ensemble:** computes a weighted average from lasso, boosting, random forest and neural net.

Implementation Pipeline: ATE Estimation (5)(6)

Objective: Strip away confounding via orthogonalization, then estimate and aggregate the final causal parameters.

(5) Orthogonalization (Partialling-Out): Using the out-of-sample ML predictions, compute the residuals $\tilde{Y}$, $\tilde{D}$, $\tilde{Z}$.

(6) Final Estimation & Aggregation: Because 2-fold cross-fitting relies on a random partition of the data, the code repeats the entire process **100 times** (100 splits).

- **Average Treatment Effect:** Calculated as the **median** of the 100 distinct causal estimates.
- **Standard Errors:** Adjusted using the **median method** to account for the estimation variability.

Results

Result 1: Replicating the Baseline (Limited Controls)

- **Objective:** To verify if the original OLS results were driven by functional form misspecification (e.g., assuming linearity).
- **Controls:** Uses the exact same limited set of **7 raw covariates** as Alesina et al. (2013) (e.g., agricultural suitability, tropical climate).

TABLE 1 Country-level estimates, partially linear model.

	(1) Lasso	(2) Reg. tree	(3) Boosting	(4) Forest	(5) Neural net.	(6) Ensemble	(7) Best	(8) OLS
Panel A: Female labor force participation 2000								
Plough use	−10.533	−9.715	−11.934	−11.325	−11.663	−11.611	−11.404	−12.401
	(3.167)	(2.818)	(3.120	(3.178)	(3.177)	(3.179)	(3.192)	(2.964)
Observations	165	165	165	165	165	165	165	165
Panel B: Share of firms with female ownership								
Plough use	−12.832	−11.543	−12.813	−12.566	−13.438	−13.337	−13.281	−15.241
	(4.006)	(3.680)	(3.959)	(4.482)	(3.873)	(4.172)	(4.285)	(4.06)
Observations	123	123	123	123	123	123	123	123
Panel C: Share of political positions held by women 2000								
Plough use	−5.165	−3.670	−4.295	−5.287	−4.465	−5.072	−4.971	−4.821
	(1.957)	(1.764)	(1.786)	(1.803)	(1.901)	(1.901)	(1.858)	(1.782)
Observations	144	144	144	144	144	144	144	144
Raw covariates	7	7	7	7	7	7	7	7

Note: Analysis of Table 4 (columns 1, 3, and 5) of Alesina et al. (2013) using DML. Column 8 reports the original paper results. Standard errors adjusted for variability across splits using the median method are reported for the DML estimates. Robust standard errors are reported in column 8. The number of covariates does not include the treatment variable. The regressions include the baseline historical ethnographic and contemporary controls.
Abbreviation: OLS, ordinary least squares.

Table 2 (Panel A): The "Full" Specification

- Expand the model to include **36 raw covariates**, adding country fixed effects, complex historical controls, and modern economic indicators.

TABLE 2 Country-level estimates with full set of controls.

	(1) Lasso	(2) Reg. tree	(3) Boosting	(4) Forest	(5) Neural net.	(6) Ensemble	(7) Best	(8) OLS
	Panel A: DML, Partially linear model. Outcome: Female labor force participation							
Plough use	−5.865	−5.951	−6.337	−6.023	−5.580	−6.062	−6.197	−9.234
	(4.892)	(4.056)	(4.235)	(4.265)	(4.740)	(4.433)	(4.416)	(4.301)
Observations	142	142	142	142	142	142	142	142
Raw covariates	36	36	36	36	36	36	36	30

$\Rightarrow$ With DML, the treatment effect of interest does not only become smaller but also statistically insignificant across all algorithms.

Table 2 (Panel B): DML-IV

- These IV estimates are identified from variation in historical plough adoption driven by crop suitability (plough-positive and plough-negative crops).
- The DML-IV coefficients are **larger** in magnitude than the OLS, PLM, and 2SLS estimates and most of them are significant at the 10% level. $\Rightarrow$ Standard estimates likely *underestimate* the true negative impact of the plough (bias towards zero). The deep cultural "scar" is likely larger than the observed data alone suggests.

	Lasso	**Reg. tree**	**Boosting**	**Forest**	**Neural net.**	**Ensemble**	**Best**	**2SLS**
	Panel B: DML-IV. Outcome: Female labor force participation							
Plough use	−41.742	−37.035	−38.766	−35.563	−21.882	−37.398	−35.898	−28.516
	(15.370)	(12.603)	(12.944)	(12.475)	(18.460)	(16.640)	(19.292)	(7.559)
Observations	160	160	160	160	160	160	160	160
Raw covariates	17	17	17	17	17	17	17	11

Discussion and Takeaways

Discussion & Key Takeaways

Empirical Confirmation

- CML confirms a **significant negative causal effect** of historical plough use on female labor force participation $\Rightarrow$ validates the original findings of Alesina et al. (2013).
- *Key finding:* The DML-IV estimates were actually **larger in magnitude** than the original 2SLS results, suggesting flexible ML isolates a stronger causal signal.

Discussion & Key Takeaways

Takeaway 1: Systematic Robustness

- Traditional robustness checks often rely on ad-hoc, manual variable selection.
- CML uses **data-driven algorithms** to systematically compare specifications, automatically incorporating **complex nonlinearities and interactions** that exhaust standard OLS.

Takeaway 2: Combining ML with IV

- ML flexibly handles *observed* confounders but cannot account for *unobserved* endogeneity (e.g., DML OLS estimates actually lost significance!).
- Integrating **CML + IV** overcomes both biases simultaneously, providing a powerful toolkit for applied economists.

Thank you

Questions?

Appendix

Table 3: Robustness Check of the Instrument (First Stage)

- For the IV strategy to be valid, the instrument (Crop Suitability) must strongly predict Plough Use.
- **Plough-Positive Crops** Strong, positive coefficient across all ML models. (Wheat/Barley $\Rightarrow$ Higher probability of Plough adoption).
- **Plough-Negative Crops** Weak or insignificant effect. (Maize/Sorghum $\nrightarrow$ Plough adoption).

TABLE 3 First-stage estimates.

	(1) Lasso	(2) Reg. tree	(3) Boosting	(4) Forest	(5) Neural net.	(6) Ensemble	(7) Best	(8) OLS
Plough-positive crops	0.526	0.480	0.590	0.600	0.384	0.498	0.408	0.678
	(0.164)	(0.136)	(0.151)	(0.140)	(0.153)	(0.159)	(0.157)	(0.204)
Plough-negative crops	0.161	-0.065	0.142	0.087	0.226	0.221	0.152	0.232
	(0.158)	(0.149)	(0.153)	(0.157)	(0.129)	(0.141)	(0.146)	(0.204)
Observations	160	160	160	160	160	160	160	160

Note: This table reports the results of the first-stage analysis of Table 2, panel B. For the DML analysis, the standard errors are adjusted for variability across splits using the median method. Robust standard errors are reported in column 8.
Abbreviation: OLS, ordinary least squares.